HR EMPLOYEE ATTRITION ANALYTICS

A SQL, Python & Power BI Project Documentation

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1,470	21.6%	3.1x	0-1 yr
Total Employees Analyzed	Overall Attrition Rate	Overtime vs Non-Overtime Risk	Highest-Risk Tenure Group

1. Project Overview

Employee attrition — when employees voluntarily leave a company — is a costly, recurring problem for organizations. Every departure carries real costs: recruiting, onboarding, training, and lost productivity while a replacement gets up to speed. This project analyzes an HR employee dataset to identify which factors are most strongly associated with attrition, and translates those findings into concrete, actionable retention recommendations.

The goal was to practice and demonstrate the complete Data Analyst workflow end-to-end: pulling and aggregating data with SQL, validating patterns statistically with Python, and communicating results through an interactive Power BI dashboard — the same pipeline used by working analysts.

2. Dataset

- **Source:** 1,470 employee records across 3 departments (Research & Development, Sales, Human Resources)
- **Fields:** Employee attributes including department, job role, monthly income, overtime status, job satisfaction, work-life balance, tenure (years at company), years since last promotion, and attrition status (Yes/No)
- **Data Quality:** No missing values or duplicate rows found during data quality checks

3. Tools & Technologies

- **SQL:** MySQL Workbench — for data storage, aggregation queries, and calculating attrition rates across dimensions
- **Python:** Pandas for data cleaning and quality checks; correlation analysis to statistically validate attrition drivers
- **Power BI:** Interactive dashboard with KPI cards, DAX measures, and cross-filterable charts for stakeholder reporting
- **GitHub:** Public repository hosting all project files, code, and documentation

4. Methodology

4.1 SQL Analysis (MySQL)

The dataset was imported into a MySQL database as a single table (employees). Aggregation queries using GROUP BY and CASE WHEN logic were used to calculate attrition rates across multiple dimensions.

```sql
-- Attrition rate by department
SELECT Department, COUNT(*) AS headcount,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes'
    THEN 1 ELSE 0 END) / COUNT(*), 1) AS attrition_rate_pct
FROM employees
GROUP BY Department
ORDER BY attrition_rate_pct DESC;
```

Sample SQL query used to calculate department-level attrition rates.

4.2 Python Exploratory Data Analysis

Using Pandas, the dataset was checked for missing values and duplicates, then the Attrition column was converted to a binary flag (1 = Yes, 0 = No) to enable correlation analysis against numeric fields such as job satisfaction, income, tenure, and age.

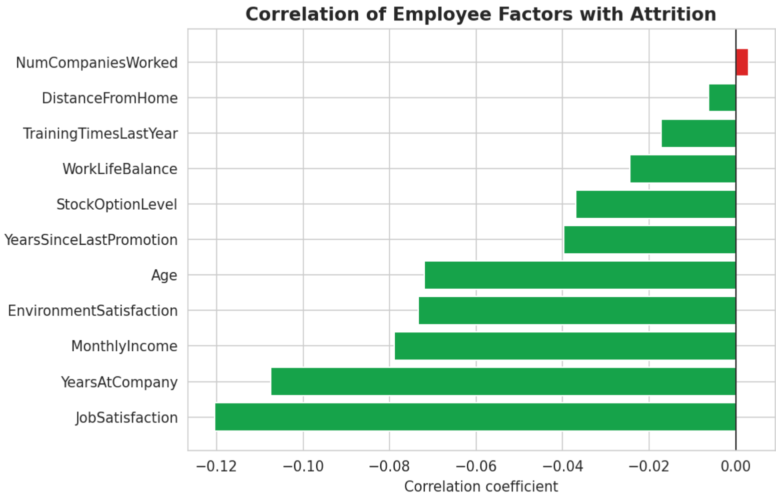

Figure 1: Correlation of employee factors with attrition. Job satisfaction, tenure, and income showed the strongest (negative) relationships.

4.3 Power BI Dashboard

Findings were translated into an interactive Power BI dashboard including two KPI cards (Total Headcount, Attrition Rate via a custom DAX measure) and three charts: attrition by department, attrition by overtime status, and an attrition trend across tenure.

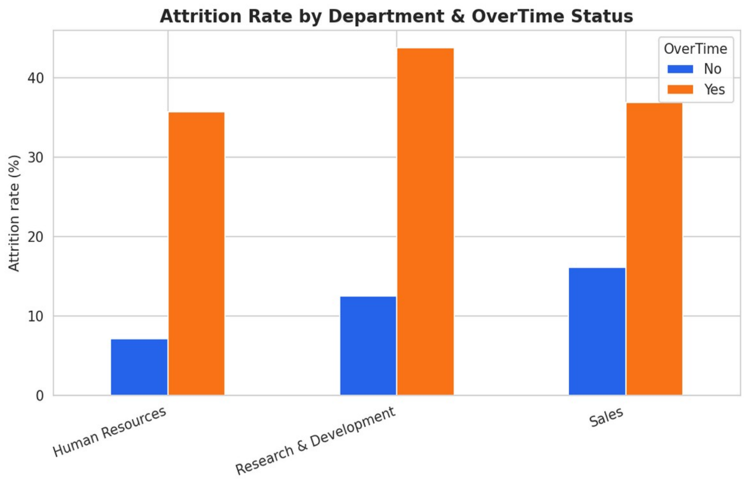

Figure 2: Attrition rate by department and overtime status.

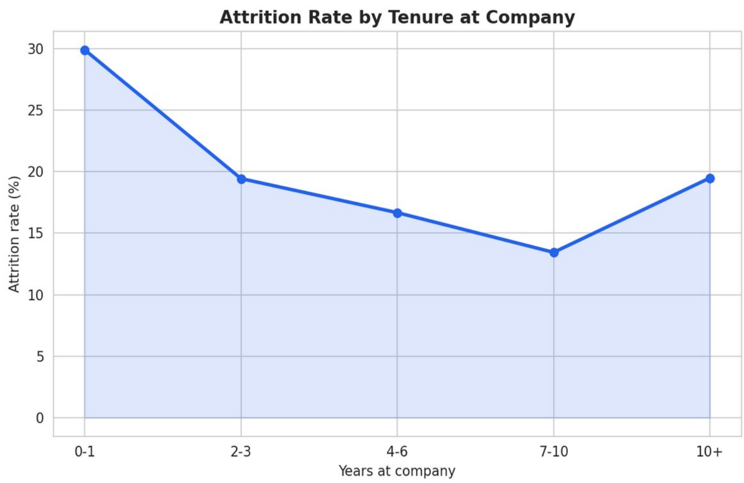

Figure 3: Attrition rate declines sharply after the first 1-2 years at the company.

5. Key Findings

- **Overtime is the strongest driver of attrition.** Employees working overtime show a substantially higher attrition rate (~41.5%) compared to those who don't (~13.5%) — roughly 3x the risk.
- **Sales carries elevated risk.** Despite having fewer total employees than R&D, Sales shows a disproportionately high attrition rate relative to its headcount.
- **Attrition is front-loaded by tenure.** Attrition peaks in the first 1-2 years (~29.9%) and drops sharply as tenure increases, flattening out after year 4.
- **Lower income bands show elevated risk.** Employees in the lowest income band show notably higher attrition than higher-paid peers, suggesting a link between compensation and retention.

6. Business Recommendations

1. Review workload distribution in overtime-heavy teams. The clear gap between overtime and non-overtime attrition suggests burnout may be driving exits — worth investigating staffing levels before it worsens.
2. Investigate Sales-specific retention drivers. Given its disproportionate attrition rate, exit interviews or manager reviews within Sales could reveal whether commission structure, quota pressure, or management style is a contributing factor.
3. Strengthen early-tenure onboarding. Since most attrition happens within the first two years, a structured onboarding program with 90-day check-ins and mentorship could catch disengagement before employees decide to leave.

7. Conclusion

This project demonstrates a complete, self-directed analytics workflow — from raw data to a stakeholder-ready dashboard and actionable business recommendations — using tools directly relevant to Data Analyst and Business Analyst roles (SQL, Python, Power BI). All findings were derived directly from the dataset without fabricated numbers, and recommendations were written the way an analyst would present them to a real HR stakeholder.

Project Repository

github.com/MattewadaCherishma/hr_attrition_analytics